

Zenmap Scanning Lab

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These target IP addresses were meant to be pinged in the lab. I couldn't reach them so decided to substitute them in with 8.8.8.8 and 1.1.1.1 which are Google and CloudFlare respectively.

Metasploitable 2 Linux Systems:

- 172.15.1.113
- 172.15.1.101
- 172.15.1.104

Windows XP Systems:

- 172.15.1.105
- 172.15.1.100
- 172.15.1.107

Step 1: Open Zenmap on Windows

Open Zenmap from the Start menu.

In the Target field, enter:

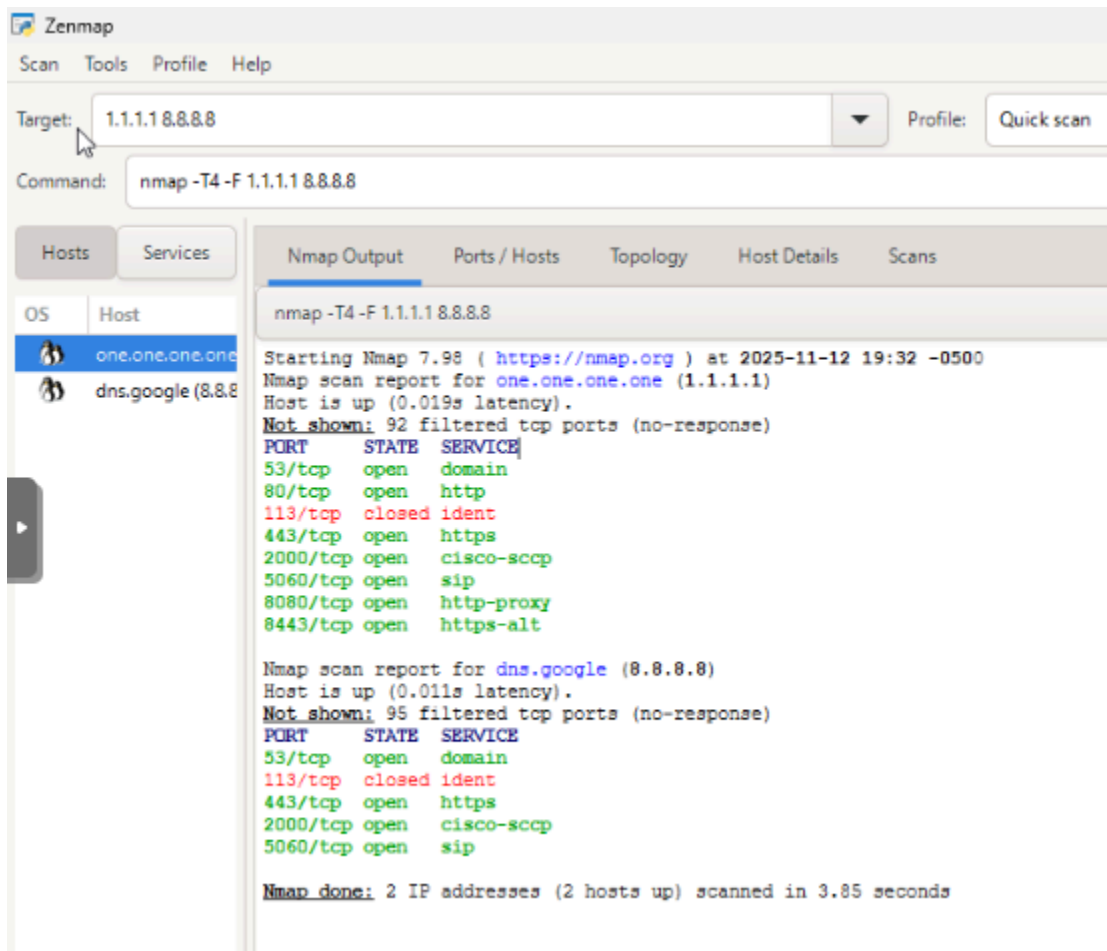
1.1.1.1 8.8.8.8

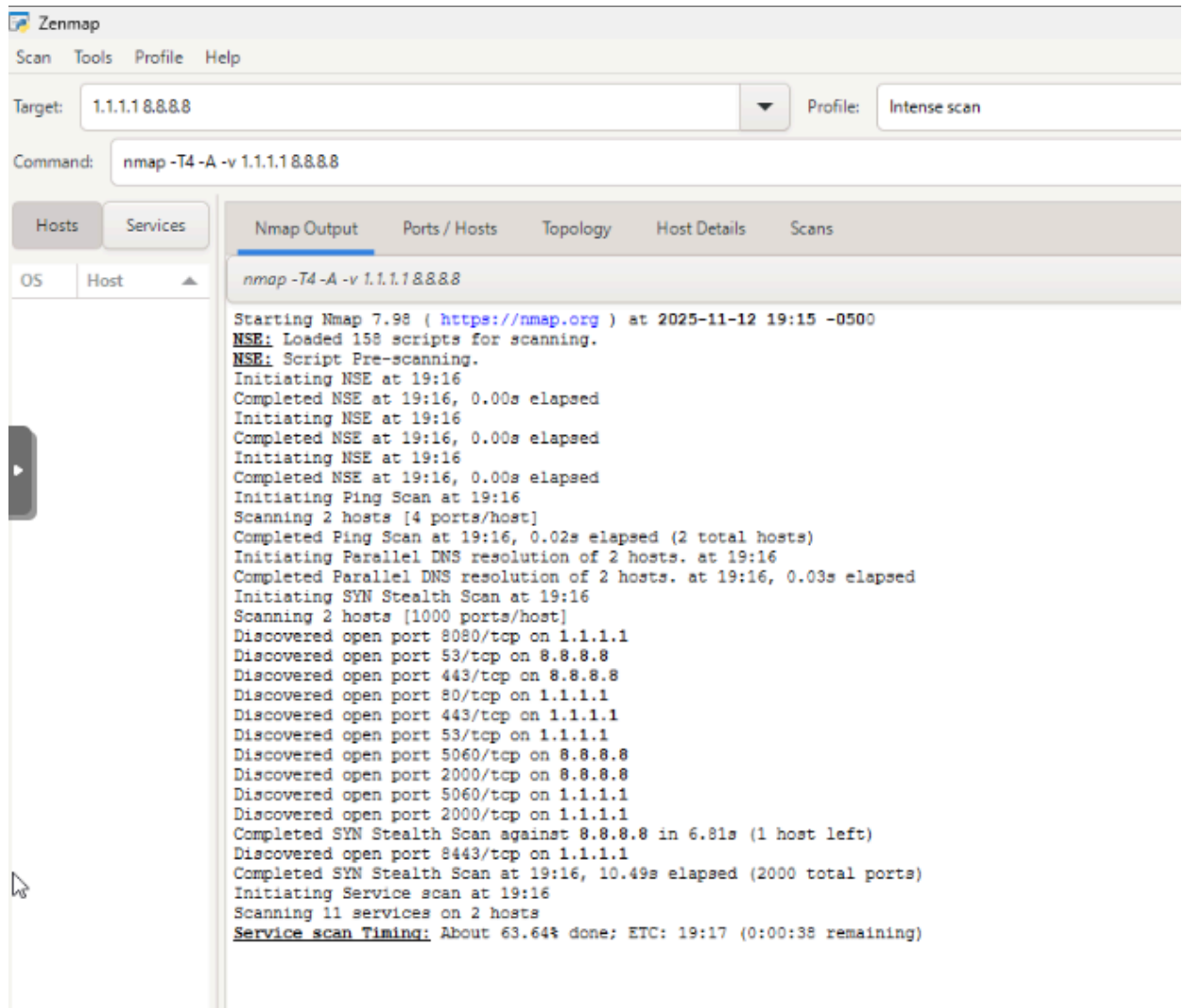
Select the scan type:

- Quick scan (for a basic check).
- Intense scan (for more details).
- Intense scan, all TCP ports (for a full analysis).
- Intense scan, all TCP and UDP ports (for advanced users).

Step 2: Perform a Basic Scan

1. Select "Quick scan" and click "Scan".





2. Observe the results, which will include host discovery, open ports, and services running.

Step 3: Perform an Intense Scan

Select "Intense scan" and click "Scan".

Zenmap

Scan Tools Profile Help

Target: 1.1.1.8.8.8.8 Profile: Intense scan, all TCP ports

Command: nmap -p 1-65535 -T4 -A -v 1.1.1.8.8.8.8

Hosts Services

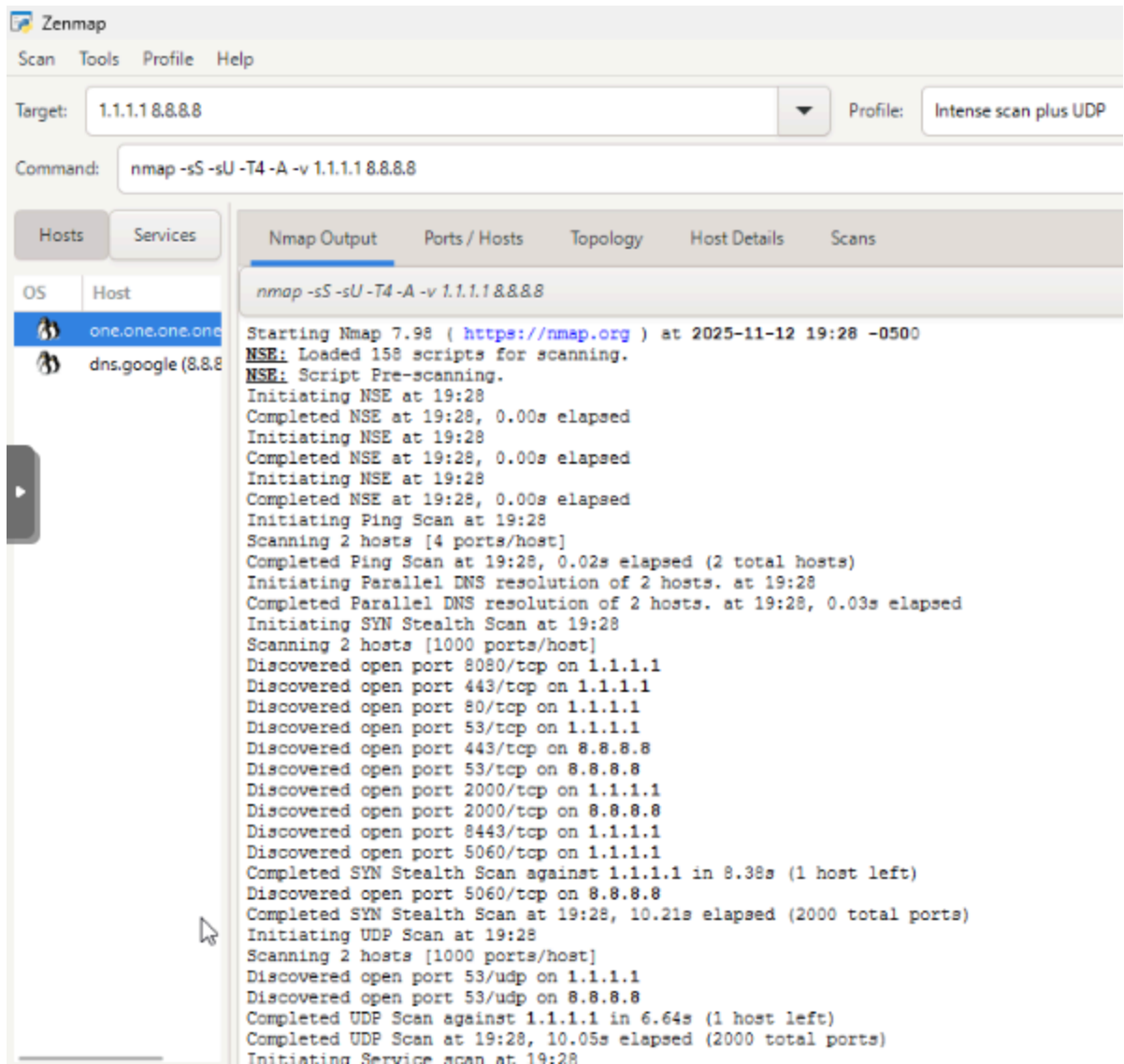
OS Host

one.one.one.one
dns.google (8.8.8.8)

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -p 1-65535 -T4 -A -v 1.1.1.8.8.8.8

Starting Nmap 7.98 (<https://nmap.org>) at 2025-11-12 19:28 -0500
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 19:28
Completed NSE at 19:28, 0.00s elapsed
Initiating NSE at 19:28
Completed NSE at 19:28, 0.00s elapsed
Initiating NSE at 19:28
Completed NSE at 19:28, 0.00s elapsed
Initiating Ping Scan at 19:28
Scanning 2 hosts [4 ports/host]
Completed Ping Scan at 19:28, 0.03s elapsed (2 total hosts)
Initiating Parallel DNS resolution of 2 hosts. at 19:28
Completed Parallel DNS resolution of 2 hosts. at 19:28, 0.03s elapsed
Initiating SYN Stealth Scan at 19:28
Scanning 2 hosts [65535 ports/host]
Discovered open port 443/tcp on 8.8.8.8
Discovered open port 443/tcp on 1.1.1.1
Discovered open port 53/tcp on 8.8.8.8
Discovered open port 80/tcp on 1.1.1.1
Discovered open port 8080/tcp on 1.1.1.1
Discovered open port 53/tcp on 1.1.1.1



Review the additional details, such as OS detection, service versions, and possible vulnerabilities.

Step 4: Analyze Results and Answer the Questions

Questions for 1.1.1.1

1. Which ports are open on 1.1.1.1?

The tcp ports 80, 443, 53, 8080, 2087, 5060, 2000, 8443, 2052, 2053,

2082, 2083, 2086, 2087, 2095, 2096, 8880, 113, and 5060 are open, and the udp port of 53.

2. What services are running on 1.1.1.1?

Cisco-sccp, domain, http, https, ident, and sip.

3. Is SSH enabled on 1.1.1.1? If so, which port?

No, SSH is not enabled on any port.

4. What version of Apache is running on 1.1.1.1?

Apache is not running on 1.1.1.1.

5. Identify any FTP service running. Which authentication method does it support?

There is no FTP services running.

6. Is there a MySQL database running? If yes, on which system and port?

There is no MySQL database running.

7. What vulnerabilities are associated with any detected RPC services?

There no RPC services detected, or any vulnerabilities associated with them.

Questions for 8.8.8.8

8. How many open ports are detected on 8.8.8.8?

The tcp ports 2000, 53, 853, 443, 5060 are detected, as well as the udp port 53.

9. Does 8.8.8.8 have SMB enabled? If yes, which ports?

8.8.8.8 does not have SMB enabled.

10. What version of NetBIOS is running on 8.8.8.8?

NetBIOS is not running on 8.8.8.8.

11. Which ports show a high-security risk?

The ports that show the highest risks are the open and exposed 5060 tcp port for SIP and the 53 UDP/TCP dns service.

12. Is there any evidence of Remote Desktop Protocol (RDP) being enabled? If yes, which system and port?

There is no evidence of RDP being enabled.

13. Are there any open Telnet services? If so, which system and port?

There are no open Telnet services.

14. Identify any open SNMP services. What potential risk does it pose?

There are no open SNMP services.

15. Are there any outdated software versions detected? If so, which?

There are no outdated software versions detected.

16. Based on the scan, which system appears most vulnerable? Why?

Report:

The scans conducted show a few worrying things that should be fixed.

1.1.1.1 has a ton of open TCP ports (80, 443, 53, 8080, 5060, 2000, 8443 and lots of other ports) plus UDP 53, and is running services like

HTTP/HTTPS, DNS, SIP, ident, and Cisco-SCCP, which makes it have a big attack surface for web exploits, DNS abuse, SIP fraud, and possible

unauthorized access to management interfaces. 8.8.8.8 has fewer ports but still exposes SIP and DNS, both of which are high-risk for signaling attacks and DNS-based abuse. There was no evidence of SMB, RDP, Telnet,

SNMP, FTP, SSH, MySQL, or NetBIOS, and no outdated software versions

showed up in this scan. To clean this up, ports not used should be closed or blocked and required services should be limited to specific IPs or a VPN/management network. What is left should be hardened with TLS and rate-limiting should be enabled, as well as turning off legacy things like ident, and segment the network so public services can't touch internal systems directly.